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Part-A. - Theory:-

1) What is a variable?

A Variable is a Container Used to Store a Value.

There are three types:

1) Var → Old way to declaration variable
2) let → Value Can be Changed.
3) Const → Value Can't be changed & Reassigned.

2) What are Javascript Data types?

* It is high level Language of Java Script.
* Java Script is a Open Source interesprinted Dianonically type Dnogramming Language.

Data types:-

Data types tell us what kind of value a variable Contains

Types:- ⇒ There are two types.

1) String.
2) Numbers
3) Boolean.
4) Undefine.
5) Null.

① Primitive
② non Primitive.

3) Difference Between == and ===

==:-

Equal Value

Ex1:-

Console.log(10==10);

output:

True.

Ex2:

Console.log(10=="10");

===:-

Equal Value and types.

Ex1-

Console.log(10===10);

True.

Ex2:-

Console.log(10==="10");

false.

4) Operators :-

1) Arithmetic Operators:

Used for mathematical operations

Ex:-

```
Console.log(a+b);
Console.log(a-b);
Console.log(a*b);
Console.log(a/b);
Console.log(a%b);
```

(+) Addition
(-) Subtraction
(*) Multification
(/) Division
(%) Remainder (or) modulo.

2) Assignment Operators :-

Used to assign or update values.

Examples

= Assign
+ = Add and assign
- = Subtract and assign
* = Multiply and assign
/ = Division and assign.
%= Modulo and assign
** = Exponentiation and assign

x+=y ⇒ x = x+y.
x-=y ⇒ x = x-y.
x*=y ⇒ x = x*y.
x/=y ⇒ x = x/y.
x%=y ⇒ x = x%y.
x**=y ⇒ x = x**y.

3) Comparison Operators :-

Used to Compare values.

Examples:

< → less than.
> → Greater than.
<= → Less than or equal.
>= → Greater than or equal.
== → Equal.
=== → Equal value and types

!= → Not Equal.
!== → Not equal value or type.

4) Logical Operators:-

Examples:

And && -> Both Conditions must be true.

Ex:

```
let age = 20;
Console.log(age >= 18 && age <= 30);
```

output :-

true.

OR || -> Atleast One Condition is true:

Ex:

```
let age = 20;
Console.log(age >= 18 || age >= 30);
```

NOT ! -> Change true to false (and) false to true

Ex:-

```
let isStudent = true;
Console.log(!isStudent);
```

output :-

false

5) Array of Objects :-

i) Array :-

Array is used to store multiple values in a single Variable

Ex:

```
let fruits = ["Apple", "Mango", "Orange", "Grapes"];
Console.log(fruits);
```

output :-

["Apple", "Mango", "Orange", "Graphs"].

=> Fruits is the array.

=> "Apple", "Mango", "Orange", "Graphs" are the Values.

=> Each Value has an index staring from 0.

0 -> Apple.
1 -> mango.
2 -> Orange
3 -> Graphs.

Object:-

An Object is used to store data in key-value pairs.

Ex:-

```
let student ={
    name : "Catherine"
    age : 23.
  Course : "Full Stack"
};
Console.log(student);
```

Output :-

```
name : "Catherine"
age : 23
Course : Full Stack
```

=> Name, age, Course -> are keys.

=> 'Catherine', 23, "Full Stack" -> are Values.

Array of Object :-

An Array of objects is an array that Contains multiple objects.

We use it to store details of multiple Students, Employ, Product and etc.

Ex:-

```
let student = [
  {name : "Cathrin", mark : 90},
  { Name : "Priya", mark : 100},
  { name : "Angel", Mark : 100},
];
```

=> Students is the array Name.

=> Each {} represents One object.

=> name and mark are Object Properties.

=> The array Contains three Student Objects.

b) What is a Function?

A function is a block of code used to perform a particular task.

We can create a function once and use it multiple times whenever needed.

1) Simple Function Example:-

```
function greet (){.
   Consde.log ("Hello Catherine");
}
 greet();
```

Output:-

Hello Catherine

function -> Keyword used to create a function.

greet -> Function name.

() -> Parentheses for parameter.

{} -> Function body, where we write the code.

2) Function with Parameters:-

Parameters are variables written inside the function Parentheses.

They recieve when the function is called.

```
function add (a,b){.
   Console.log (a+b);
}
 add (10,20);
```

output:-

30.

=> a and b are parameters.

=> 10, 20 are arguments.

=> a+b add the two numbers.

=> Console.log() prints the results.

3) Function with Return :-

Return is used to send a result back from a function.

Ex:

```
function add(a,b){
    return a+b;
}
let result = add(10,20);
console.log(result);
```

=> return a+b send the result back.

=> result stores the returned value

=> Console.log(result) Print 30.

O/P//

30.

Ex

Parameters -> Variable in function definition — a, b

Arugument -> Value passed to function — 10, 20.

Return -> Sends the result back — return a+b.

Part - B Basic Programs :-

7) Even Or Odd :-

Even Number :-

A number that is exactly divisible by 2. The remainder 0.

Ex :-

[2, 4, 6, 8, 10, 12]

Odd Number :-

A number that is not exactly divisible by 2. The remainder is 1.

Ex :-

[1, 3, 5, 7, 9]

Example1:-

```
let number = 27;
if (number % 2 == 0){
  Console.log("Even number");
}
else{
  Console.log("odd number");
}
```

Output → Odd Numbers.

The Condition is false, the else block executes and print odd numbers.

Example 2:-

```
let number = 20;
if (number % 2 == 0){
  Console.log("Even no");
}
else{
  Console.log("odd no");
}
```

Out put => Even Numbers.

8) For loop:-

A for loop is used to repeat a block of code multiple time Until a Condition becomes false.

Ex:

```
for (let i = 1; i <= 50; i++){
  Console.log(i);
}
```

=> let i = 1 loop starts from 1.

=> i <= 50 loop runs Until 50.

=> i++ Increases the value of i by 1 each time.

=> Console.log(i) Prints the current numbers.

Out put :-

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50.

Even numbers:-

To find even number we use the modulus %. Operator.

Ex:

```
for (let i=0; i<=50; i++){
    if (i%2==0){
        Console.log(i);
    }
}
```

Output:-

2, 4, 6, 8, 10, ... 50.

9: Factorial -> Using for loop:-

Factorial means multiplying a number by all the positive numbers before it.

It is represented by !.

Examples:

5! = 5 x 4 x 3 x 2 x 1.

Out put:

120.

number = 5 -> Given number.

factorial = 1 -> Storing value.

for loop -> repeats from 1 to 5.

factorial * i -> multiplies each number.

final Out put -> 120.

5 x 4 = 20.

20 x 3 = 60

60 x 2 = 120.

120 x 1 = 120.

Examples:

```
let number = 5;
let factorial = 1;
for (let i=1; i<=number; i++){
    factorial = factorial * i;
}
Console.log(factorial);
```

Out put:-

120.

10. Fibonacci Series :-

Fibonacci Series is the Next number equal to the Previous number in add current number.

Next Number = Previous number + Current number.

for loop :-

```
let a=0;
let b=1;
for (let i=1; i<=8; i++){
   Console.log(a);
let c=a+b;
  a=b;
  b=c;
}
```

a=0; b=1;

=> 0+0 =0

=> 0+1 =1

=> 1+1 =2.

=> 2+1 =3.

=> 3+2 =5

=> 5+3 =8

=> 8+5 = 13.

Output :-

0, 1, 1, 2, 3, 5, 8, 13.

Part C - Array & Object :-

11) Array :-

```
let number = [10, 15, 20, 25, 30, 35, 40];
for (let i =0; i <numbes.length ; i++){
  if (number[i] % 2 == 0){
    Console.log(number[i]);
  }
}.
```

Output :-

10, 20, 30, 40.

i = 0 => First Position.

numbers.length -> Total number of elements.

i++ -> moves to the next elements.

numers -> [10, 15, 20, 25, 30, 35, 40].

number [0] -> 10.
numbers [1] -> 15.
number [2] -> 20
number [3] -> 25
number [4] -> 30.
number [5] -> 35
number [6] -> 40.

If (numbers[i] % 2 === 0)

10 % 2 = 0 -> Even no.
15 % 2 = 1 -> Odd no.
20 % 2 = 2 -> Even no.
25 % 2 => 3 -> Odd no.
30 % 2 = 4 -> Even no.
35 % 2 = 5 -> Odd no.
40 % 2 = 6 -> Even no.

12) Array Total :-

```
let marks = [80, 75, 90, 85, 70];
let total = 0;
for (let i=0; i < marks.length; i++){
    total = total + marks[i];
}
Console.log (total);
```

80
75
90
85
70
400

Output :-

400.

13) Object :-

```
let student = {
  name : "Arun",
  age  : 21,
  Course : "Java Script",
  mark : 85.
};
Console.log(student);
```

Output :-

name -> Arun.
age -> 21.
Course -> Java Script.
mark -> 85.

Step 2 :-

```
Console.log(student.name);
Console.log(student.age);
Console.log(student.course);
Console.log(Student.mark)
```

Output :-

Arun.
21
Java Script.
85

Ex :-

```
let Student = {
   name : "Arun",
   age : 21,
  Course : "Java Script",
   mark : 85.
};
Console.log (Student.name);
Console.log (Student.Course);
Console.log(Student.mark);
```

Output :-

Arun, Java Script, 85.

4) Array of objects :-

```
let students = [
 { name : "Arun", mark : 80 },
 { name : "Kumar", mark : 45 },
 { name : "Priya", mark : 90 },
 { name : "Ram", mark : 35 }
];

for( let i=0; i<students.length; i++){
  if (student [i] . mark >= 50){
  Console.log(students [i] . name);
}.
}.
```

Out put :

Arun.

Priya.

Arun → 80 >= 50. → true.

Kumar → 45 >= 50 → False.

Priya → 90 >= 50 → true.

Ram → 35 >= 50 → False.

Part - D

15. Function & Logic :-

Function - Largest Number :-

```
function number =>a = 50, b = 100;

function LargestNumber(a,b){
    if (a>b){
     return (a);
   }
   else {
     return (b);
   }
  }
Console.log(Largest number (50, 100));
```

Output :-

100.

function → Reusable of code.

Parameter → a, b.

Argument → 50, 100.

Return => gives the Answer.